

Salman Abdullah Alsalman

Computer Science Lecturer | Artificial Intelligence & Computer Vision

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Academic Profile

Computer science lecturer with interests in artificial intelligence, machine learning, deep learning, and computer vision. Research experience spans medical image analysis, remote sensing, and applied machine learning.

Education

University of Kentucky, United States 2022–2023

Doctoral studies in Computer Science (degree not completed) - GPA: 3.667/4.00.

Selected coursework: Biomedical Imaging, Computer Graphics, and Algorithm Design.

San Diego State University, United States 2016–2019

M.S. in Computer Science (coursework-based) - GPA: 3.07/4.00.

Selected coursework: Applied Computer Vision, Machine Learning, and Data Mining.

Imam Muhammad Bin Saud Islamic University, Saudi Arabia 2012

B.Sc. in Computer Science, Second Class Honor Degree - GPA: 4.67/5.00.

Publications

- [1] **Salman Alsalman** and Amina Salhi. HCFFormer: a hybrid CNN–transformer framework for heterogeneous remote sensing image change detection. *International Journal of Remote Sensing*, 47(16), 7054–7117, 2026. DOI: 10.1080/01431161.2026.2691979.
- [2] Biru Rajak, **Salman Alsalman**, Maksudjon Usmonov, Rohit Markan, G. Sajiv, and M. Ponni Valavan. Energy-Efficient Ubiquitous Sensor Collaboration System Enhancing Real-Time Decision Making in IoT Ecosystems. *2025 IEEE 2nd International Conference on Green Industrial Electronics and Sustainable Technologies (GUEST)*, 2025. DOI: 10.1109/GUEST66547.2025.11387726.
- [3] Srigouri Kosuri, Abdurasul Bobonazarov, Shazia Ali, **Salman Alsalman**, S. Suma Christal Mary, and R. Dhaaraani. Modeling Employee Emotional Intelligence Progression Using Self-Supervised Deep Learning and Longitudinal Organizational Behavior Datasets. *2025 IEEE Pune Section International Conference (PuneCon)*, 2025. DOI: 10.1109/PuneCon67554.2025.11377991.
- [4] Ahmed Biyabani, **Salman A. Al-Salman**, and Khalaf S. Alkhalaf. Embedded real-time bilingual ALPR. *2015 International Conference on Communications, Signal Processing, and their Applications (ICCSPA)*, Sharjah, 2015. DOI: 10.1109/ICCSPA.2015.7081311.

Selected Academic Projects (during graduate studies)

- **Malaria Cell Detection using Faster R-CNN and Curriculum Learning** 2023
Explored Faster R-CNN and curriculum learning using Detectron2 and the M5-Malaria dataset. Converted annotations to COCO format and evaluated detection across different microscopes and magnifications.
- **Analysis of Traffic-Citation Bias in San Diego** 2019
Analysed San Diego traffic-stop records from the Stanford Open Policing Project. Applied data cleaning and feature selection, comparing neural networks, decision trees, and random forests to examine patterns in ticket issuance.
- **Liver Disease Prediction using Classification Algorithms** 2017
Compared seven classifiers on the Indian Liver Patient Dataset using Python and scikit-learn. Preprocessed patient biomarker data and evaluated models using accuracy, confusion matrices, and execution time.
- **Predicting Safety Risks from Street View Photos** 2017
Explored crime-risk classification using street-view imagery and crime records from five U.S. cities. Combined image collection and cleaning, spatial clustering, and support vector machine classification.

Academic and Teaching Experience

Prince Sattam bin Abdulaziz University, Department of Computer Science

Lecturer

Jan 2020–Present

Teaching Assistant

Nov 2012–Dec 2019

- Teach computer science courses, including Discrete Mathematics, Data Structures, Programming, and Web Applications Development.
- Support students through academic advising and final year project supervision.
- Contribute to departmental committees and research collaborations.

Tuwaiq Academy

Jan–Feb 2024

AI Instructor, SDAIA GDP Program (Contractor)

- Delivered Python, neural-network, and large-language-model teaching sessions.
- Guided practical projects and supported Python certification preparation for the students.

Additional Professional Experience

- **Smartech Company — IT Consultant (Contractor)** Aug 2020–Jul 2021
Managed IT projects focused on healthcare application development and advised on strategic decisions. Coordinated communication among developers, designers, and clients throughout project delivery.
- **Elm Company — Software Developer (Intern)** Jul–Sep 2011
Analysed, designed, and developed a J2EE content-management system for Elm’s Innovation Unit.
- **IMAMU — Computer Lab Assistant (Part-Time)** Sep 2010–Aug 2011
Assisted students in the College of Computer Science and Information Systems with programming problems. Delivered supplementary lectures and practical programming guidance.
- **Saudi Aramco — IT Assistant (Intern)** Jul–Sep 2009
Provided on-site and remote IT support, troubleshooting user issues and setting up computer and network equipment.

Methods and Tools

- **Computer vision and machine learning:** Object detection, image classification, CNN–Transformer modeling, curriculum learning.
- **Programming and tools:** Python, Java, MATLAB, C++, pandas, NumPy, scikit-learn, Detectron2, Git, Jupyter Notebook, Docker.

Certifications and Professional Development

- **IBM AI Skills Accelerator Program**, IBM SkillsBuild Apr–Jun 2026
- **Certified Artificial Intelligence Consultant (CAIC)** Mar 2025
United States Artificial Intelligence Institute | Credential: 8765319297
- **McKinsey Forward Program** Jan–Apr 2024

Awards and Distinctions

- **First Place and Best Science Graduation Project**, CCIS, IMAMU 2012
Title: *License Plate Recognition System*.
- **Mawhiba Gifted Students Summer Program**, IPA 2004
Nominated for and completed a two-month KACST–Mawhiba program focused on database management systems at the Institute of Public Administration, Riyadh.

Languages

Arabic: native | **English:** full professional proficiency.